CHECKREALITY AI: A MULTIMODAL DEEPFAKE AND MISINFORMATION DETECTION SYSTEM FOR DIGITAL CONTENT AUTHENTICITY

PROJECT SYNOPSIS

OF MAJOR PROJECT

BACHELOR OF TECHNOLOGY

Computer Science Engineering

SUBMITTED BY

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Baba Banda Singh Bahadur Engineering College

Fatehgarh Sahib

July-Dec, 2026

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1. INTRODUCTION

The creation, manipulation, and dissemination of digital content by **AI** have been transformed by recent developments in **Generative AI (GenAI)**, **Large Language Models (LLMs)**, **Generative Adversarial Networks (GANs)**, **Diffusion Models**, and **Transformer based** architectures. These technologies can produce text, images, videos and speech that are very realistic looking and sounding, and they can bring great benefits to fields such as education, healthcare, entertainment, journalism, business automation and scientific research. But those same technologies have also enabled the mass production of **deepfakes** and **AI-generated misinformation**, raising significant concerns around **digital trust, cybersecurity, privacy and information integrity.**

The growing availability of powerful generative models has made it easier than ever to create manipulated digital content. **Deepfake images and videos, synthetic voices and AI-generated textual disinformation** can now be easily produced with very little technical skill and are very similar to content created by humans. This manipulated content is increasingly used for **identity theft, financial fraud, political propaganda, social engineering attacks, cybercrime, fake journalism and online harassment** and presents a burgeoning cybersecurity challenge rather than just a multimedia processing challenge in terms of verifying digital authenticity.

The rapid growth of **social media platforms**, **instant messaging applications**, and **online news portals** has further accelerated the spread of misinformation. A single manipulated post combining **fabricated text**, **edited images**, **deepfake videos**, and **synthetic speech** can reach millions of users within minutes, influencing public opinion before any manual verification can be performed. Consequently, misinformation campaigns have evolved from **single-modality attacks** to **multimodal attacks**, where multiple manipulated media types reinforce each other to increase credibility and deceive users more effectively.

The existing Artificial Intelligence based detection systems have shown promising performance in the manipulated content detection. Special models are suggested in many studies for **fake news detection, deepfake image detection, video forgery detection,** and **synthetic speech detection.** However, **most existing solutions are modality-specific**, analyzing only one type of media in isolation. Even recent multimodal methods often only focus on **text-image pairs**, while the **audio and video** modalities are insufficiently integrated. Moreover, most state-of-the-art models focus on benchmark accuracy, while they often require **large computational resources**, have **limited generalization to unseen deepfake**

generation techniques, provide **few interpretability**, and are still hard to deploy as practical web-based applications.

In recent research, we have seen the transition from isolated detection models **to multimodal Artificial Intelligence systems** that are able to learn the relations between different media modalities. Current approaches leverage **cross-modal feature interaction, attention mechanisms, multimodal transformers, contrastive representation learning, and feature-level fusion** to enhance robustness to sophisticated AI-generated misinformation. The approaches demonstrate that analyzing the semantic consistency among multiple modalities can provide a much more reliable authenticity assessment than analyzing each modality independently.

Another significant challenge that has been pointed out in recent literature is the lack of **Explainable Artificial Intelligence (XAI)** in deepfake detection systems. Most of current models **are black-box classifiers** and provide only a **Real/Fake prediction** without explaining the reason of their decisions. This limited interpretability diminishes user trust particularly in applications that involve journalism, education, digital forensics, and cybersecurity where it is often as important to understand **why** the content is considered manipulated as it is to understand the prediction itself.

Moreover, recent benchmark studies show that many deepfake detection models have bad **cross-dataset generalization**. Performance of models trained on specific datasets often degrades when evaluated on **unseen manipulation techniques, different compression rates, novel generative models or real-world social media content**. This underscores the necessity of detection systems focusing on **robustness, generalization** and **deployment** in the real world, rather than just maximizing the accuracy of a benchmark.

To tackle these challenges, this project proposes **"CheckReality AI: A Multimodal Deepfake and Misinformation Detection System for Digital Content Authenticity"**. The system's objective is to build a **holistic web-based platform** able to evaluate **textual, visual, video and audio content** through dedicated Artificial Intelligence models. The system will not just make independent predictions, but will use a **multimodal fusion framework** to fuse the feature representations of each modality to produce a holistic **authenticity assessment, confidence score, and explainable analysis report.** This design enables more reliable detection of complex multimodal misinformation campaigns, while improving transparency and user confidence.

The proposed architecture combines concepts from **Natural Language Processing (NLP), Computer Vision, Audio Signal Processing, Deep Learning, Multimodal Machine Learning, Explainable AI, and Cybersecurity** into a single modular framework. The system is designed to enable the future addition of new deepfake detection models and to allow for scalability for deployment as a **web application, REST API, browser extension or Software-as-a-Service (SaaS)** platform. The research proposed here will contribute to the development of next-generation digital content authenticity verification systems that can address the rapidly changing environment of AI-generated misinformation, with a focus on **multimodal reasoning, confidence-aware decision-making, explainability and practical deployment.**

2. LITERATURE SURVEY

The proliferation of **Generative Artificial Intelligence (Generative AI)** has revolutionized the digital information ecosystem by facilitating the generation of highly **realistic AI-generated text, deepfake images, synthetic videos and voice-cloned audio.** These tools have turbocharged innovation in education, health care, journalism, entertainment and digital communication but also turbocharged the spread of misinformation and manipulated media. Therefore, the digital content authenticity has been a hot research topic in **Artificial Intelligence (AI), Computer Vision (CV), Natural Language Processing (NLP), Multimedia Forensics and Cyber Security.** Recent researches reveal a clear trend to shift from **single-modality detection** to **multimodal learning,** where complementary information from multiple media sources is jointly analyzed to improve the detection accuracy, robustness and Reliability **[1]–[4].**

The first efforts in this area concentrated on **multimodal fake news detection** that relies on text and visual information. Alonso-Bartolome and Segura-Bedmar **[1]** showed that combining textual semantics with image representations can greatly enhance fake news classification compared to unimodal approaches, suggesting that language and visual context are complementary. Extending this work, Giachanou *et al*. [2] proposed a **multimodal multi-image detection framework** that exploits the semantic relationships among multiple images associated with the same news article, resulting in better contextual understanding and classification performance. These studies emphasized the value of multimodal learning but were largely limited to **text-image analysis**, which is less applicable to modern misinformation campaigns that increasingly involve manipulation of **audio and video.**

With the development of multimodal learning, it has evolved from simple feature concatenation to **cross-modal representation learning** in which the semantic interaction between different modalities is learned automatically. Multi-grained information fusion methods **[3]** introduced hierarchical fusion strategies to capture the relationships at different levels of representations, enabling richer feature interaction compared to traditional late fusion approaches. Similarly, **self-learning multimodal architectures [5]** reduced the need for manually designed fusion rules by automatically discovering complementary relations among heterogeneous media. Attention-based **text-image fusion models [6]** further improve semantic alignment by dynamically attending to informative features during prediction. These methods have significantly improved multimodal representation learning but mainly focused on **text-image consistency** with **video and audio** modalities still relatively ignored.

Cross-modal alignment is a large step forward for multimodal misinformation detection to bridge the semantic gap between different data sources. Cross-modal learning allows the models to learn shared

representations so as to capture the semantic consistency across different media types, rather than treating modalities independently. Liu *et al*. **[7]** found that aligning textual and visual representations can significantly improve fake news detection by enhancing the interactions between modality-specific features. Compared to conventional fusion strategies where the features are fused after independent learning, cross-modal alignment allows semantic interaction during the feature extraction, resulting in more discriminative multimodal representations. However, most of the existing alignment methods are still limited to **text-image relations**. Extending these methods to model **text, image, video and audio** in a unified manner is still a major research challenge.

With the increasing interest in multimodal misinformation detection, a number of **survey studies** have systematically reviewed existing methods, benchmark datasets, and emerging research directions. Hangloo and Arora **[4]** provided a survey of multimodal fake news detection techniques and reported that, despite significant improvements in detection accuracy, current systems still suffer from dataset dependence**, limited robustness**, and **poor integration of multimodal information**. Recent surveys **[9]–[12]** also provide further categorization of existing methods such as **early fusion, late fusion, intermediate fusion, attention-based fusion, Transformer-based fusion and cross-modal interaction learning.** Taken together these surveys point to a research shift away from the maximization of benchmark accuracy toward **generalization, computational efficiency, explainability, and deployment** in the real world. However, these advances consistently highlight the absence of lightweight and unified frameworks capable of analyzing multiple digital media modalities simultaneously.

Multimodal fake news detection primarily focuses on the semantic consistency between **text** and images. However, deepfake detection has additional challenges due to the **spatial, temporal and acoustic** inconsistencies of the manipulated contents. As opposed to fake news detection, deepfake detection requires recognizing subtle artifacts such as **facial manipulation, lip-sync inconsistencies, temporal inconsistencies, and synthetic speech.** Recent review studies **[13], [14]** show that state-of-the-art deepfake generation methods are manipulating more than one modality in the same content, which makes **multimodal detection** methods much more reliable than the unimodal ones. This evolution has encouraged researchers to design unified architectures that can simultaneously analyze heterogeneous multimedia data.

To address these issues, several **audio-visual multimodal deepfake detection frameworks** have been proposed by researchers. Salvi *et al*. **[15]** proposed a robust multimodal architecture exploiting both audio and visual information. Their model achieves better detection performance than vision-only

models, especially when visual manipulation artifacts are subtle. Based on this idea, the **FRADE (Forgery-aware Audio-distilled Multimodal Learning)** framework **[16]** utilizes **knowledge distillation** to transfer the forgery-aware information from the audio stream to assist visual feature learning. FRADE differs from traditional fusion methods that fuse independently extracted features, by encouraging interaction in representation learning, resulting in more discriminative multimodal features and better robustness.

Audiovisual representation learning proposed another approach, where the goal is to learn a shared embedding, rather than explicit fusion of modality-specific features. The **MultimodalTrace** framework **[17]** showed that learning audiovisual representations jointly effectively captures inconsistencies between facial movements and speech synchronization that are often missed by unimodal detectors. Likewise, Liu *et al.* **[18]** proposed a multimodal forgery clue enhancement framework to combine **spatial, frequency-domain** and **cross-modal** information to enlarge manipulation artifacts before classification. The methods have stronger robustness to sophisticated deepfake generation methods than previous fusion-based methods. However, the higher computational complexity and dependence on large-scale multimodal datasets still hinder the real-time deployment.

As the research has changed from algorithm development to real applications, more and more attention has been paid to **multimodal deepfake detection in real-life situations**. In view of the popularity of manipulated short-form videos on modern social media platforms, short-video deepfake detection frameworks **[19]** designed to reduce computational overhead while preserving temporal consistency analysis. Experimental studies **[20]** showed that the **multimodal feature fusion** always performs better than the independent modality-specific classifiers by exploiting the complementary information of different media sources. Based on these results, unified multimodal frameworks **[21]** that are able to process heterogeneous digital content, comprising of multiple detection modules in a single architecture. These frameworks are more flexible and more generally applicable than earlier modality-specific systems. However, their multiple pretrained models and complex processing pipelines lead to high computational costs, limiting their deployment in real-time and resource-constrained scenarios.

A major progress in recent years is the transition from **decision-level** to **feature-level multimodal fusion.** State-of-the-art architectures fuse the feature representations at intermediate learning stages, enabling a deeper semantic interaction of the modalities, while earlier systems typically combine the final predictions of single models. Hong *et al.* **[22]** showed the effectiveness of multimodal feature fusion in integrating **spatial, frequency-domain** and **semantic information**, leading to an improved robustness against sophisticated deepfake generation techniques. Similarly, Zhang *et al.* **[23]** have

shown that the fusion of handcrafted and deep learning features improves the discriminative power of the detection models. Hybrid deep learning frameworks **[24],** which combine complementary neural architectures while maintaining reasonable computational efficiency, improved the feature representation more recently. More recently, prototype-based multimodal learning **[25]** shifted the focus from dataset-specific artifacts learning to the modeling of generalized forgery patterns, resulting in more robustness against unseen manipulation methods. Despite the substantial improvement of detection performance by these methods, their structural complexity and computational cost still pose practical challenges for deployment.

Despite substantial progress in multimodal learning, **generalization** remains one of the most critical Despite the remarkable progress of multimodal learning, **generalization** is still one of the most important challenges in deepfake detection research. Many state-of-the-art models perform very well on the benchmark datasets but show a significant drop in performance when tested on **cross-domain, real-world,** or **unseen** manipulation techniques. Hence, recent research has shifted from benchmark accuracy to **robustness** and **generalization**. The **Deepfake-Eval-2024** benchmark **[26]** tests detection models on in-the-wild deepfakes and finds that many top-performing methods fail in realistic scenarios. Similarly, the **Omni-Fake benchmark [27]** encourages the development of models that can generalize beyond controlled lab datasets by providing a unified evaluation across different multimodal social media content. In support of these findings, large-scale survey studies **[28]** find **cross-dataset robustness, domain adaptation, generalization** to be challenges that current multimodal detection techniques only partially address.

With the rapid development of Large **Language Models (LLMs),** the task of verifying the authenticity of digital content has shifted from multimedia forensics to **AI-generated text detection**. The detection of AI-generated text is different from the detection of fake news, which aims to find misleading or fabricated information, as it focuses on the differentiation of **machine-generated** text from human-written text on the basis of linguistic, semantic and statistical features. Recent works **[29]** show that language models based on Transformer successfully learn subtle textual features related to the generation of synthetic content. However, comparative evaluations of the current detection tools **[30]** demonstrate inconsistent performance for different LLMs, writing styles, and prompting strategies, hinting that the current detectors are still vulnerable to the rapid evolution of text generation models. Recent review studies **[31]** emphasize the need for continuous model adaptation, benchmark updates, and **periodical retraining** to ensure reliable detection of AI-generated text in practical applications.

In addition to the improvement in detection accuracy, the importance of **Explainable Artificial Intelligence (XAI)** in enhancing the transparency and trustworthiness of multimodal authenticity verification systems has been increasingly recognized in recent works. Traditional deep learning models are generally **black-box classifiers** that only output binary **Real/Fake** predictions and provide no explanation for the evidence involved in the decision. These models often achieve high predictive performance, but their lack of interpretability limits their use in areas like **digital forensics, journalism, law enforcement** and **cybersecurity** where transparency in decision making is paramount. Thus, research has been gradually diverted to the design of systems that can provide **confidence-aware, interpretable,** and **human-understandable** predictions.

To address this problem, Jadhav *et al.* **[8]** proposed an **explainable multilingual and multimodal fake news detection framework** that combines multimodal learning and interpretable prediction mechanisms. Unlike traditional classifiers, which produce a final decision, the framework provides the textual and visual evidence behind each prediction, thus improving transparency and user trust. Momin *et al.* **[32]** similarly reviewed explainable deepfake detection techniques across multiple modalities and concluded that future detection systems should include **confidence estimation, visual attention analysis, and human-interpretable explanations** along with prediction accuracy. Explainability has been an emerging research direction, but most existing methods are still tightly coupled with individual detection models and thus of limited use in **general-purpose multimodal authenticity verification systems.**

Another research direction is to develop **integrated multimodal authenticity verification frameworks** that can evaluate different types of digital media in the same platform. Previous works have generally studied **fake news detection, deepfake image detection, video forgery detection and AI-generated text detection** as isolated problems. This resulted in fragmented solutions, requiring different processing pipelines for different media. More recent frameworks **[21], [24], [33]** have overcome this limitation by including multiple detection modules in unified architectures able to process heterogeneous digital content. Such systems are more flexible and more generally applicable than modality-specific solutions. However, they still suffer from **computational complexity, dependency on multiple pretrained models, scalability, and limited explainability**. Moreover, most of the evaluations are done on benchmark datasets and very few studies explore deployment-oriented architectures that are appropriate **for real-time web applications or resource-constrained environments.**

From the literature reviewed, it appears that there is a clear evolution in **multimodal authenticity verification.** Early works demonstrate the advantages of employing both **text and visual** information to detect fake news **[1]–[3].** Then **cross-modal interaction, attention mechanisms,** and **feature fusion** are proposed to enhance multimodal representation learning **[5]–[7].** With the emergence of more advanced deepfake generation methods, research efforts have been directed **toward audio-visual learning, multimodal feature fusion and representation learning [15]–[25].** In the meantime, benchmark studies have exposed the persistent challenges on **generalization and real-world robustness [26]–[28].** The recent surge in the use of Large Language Models has expanded the scope of authenticity verification to include detecting **AI-generated text [29]–[31]. Explainable Artificial Intelligence** has become a key element in enhancing transparency and building user trust **[8], [32].** Together, these developments show the move from detectors that are specific to a modality toward ever more **integrated, intelligent and user-centric** systems for verifying authenticity.

Despite these advances, some important problems are still unsolved. Most existing systems are still limited to a few **modality combinations**, e.g., **text-image or audio-video**, but not **text, images, videos, and audio in a single system.** Moreover, many methods optimize for **benchmark performance** at the expense of **cross-dataset robustness, computational efficiency, and deployability,** limiting their practical use. While recent works have dealt with the problems of **multimodal feature fusion and explainability**, a lot of these models still require huge computational resources and provide limited support for **confidence-aware decision-making** in real world environments.

The proposed **CheckReality AI: A Multimodal Deepfake and Misinformation Detection System for Digital Content Authenticity** aims to address these limitations by developing a **unified multimodal framework** that integrates **modality-specific deep learning models, cross-modal feature fusion, attention-based decision-making, confidence-aware prediction, and explainable authenticity** reporting in a **single web-based platform**. The system we propose aims at offering a complete and trustworthy solution for verifying the authenticity of digital content in **text, images, videos and audio,** with a strong emphasis on **practical deployment, scalability, transparency and rich multimodal analysis.**

3. PROBLEM STATEMENT

The rapid development of **Generative Artificial Intelligence (Generative AI)** has made it possible to produce highly realistic **AI-generated text, deepfake images, manipulated videos,** and **synthetic speech**, and this presents significant challenges in validating the authenticity of digital content. Despite recent advances in multimodal detection, not all practical issues have been solved yet.

The key issues identified are:

- **AI-generated text, deepfake images, manipulated videos, and cloned voices** are increasingly used for **misinformation**, **identity theft**, **financial fraud**, **political propaganda**,
- Contemporary misinformation campaigns often combine **multiple media modalities (text, image, video, and audio)**, and this makes the task of authenticity verification much more challenging than single-media analysis.
- Current multimodal detection systems are usually limited to certain **modality combinations** (e.g., **text-image** or **audio-video**) and lack a **unified framework** that can evaluate all important types of digital content at the same time.
- Many existing detection models **generalize** poorly under **real-world**, **cross-domain**, or **unseen deepfake generation techniques**, which undermines their practical reliability.
- Most existing systems only offer a **binary prediction** of **Real/Fake** any **estimation** of **confidence** or **explainable reasoning**, making it hard for users to understand and trust the prediction results.
- Due to **high computational resources**, **educational institutions, researchers** and **small organizations** find it challenging to deploy and practically implement several state-of-the-art detection models in **real-time.**
- Many commercial digital authenticity verification platforms are **closed-source**, rely on **subscription-fees**, or offer **little transparency** into how prediction are made.
- At present, there is **no lightweight, scalable, explainable, and user-friendly web-based platform** capable of providing full **multimodal authenticity verification** for **text, images, videos, and audio** in a single integrated system.

To this end, this project proposes the development of *"CheckReality AI: A Multimodal Deepfake and Misinformation Detection System for Digital Content Authenticity"* to provide a unified, scalable, and explainable web-based platform that incorporates multimodal deep learning models, feature fusion, confidence-aware prediction, and authenticity reporting for reliable verification of digital content.

4. RESEARCH GAP

The literature review shows that there has been a significant progress in **multimodal fake news detection**, **cross-modal learning**, **audio-visual deepfake detection**, and **feature fusion techniques**. However, there remain several important gaps in the research that limit the practical applicability of current systems.

The key research gaps identified are:

- The vast majority of existing multimodal systems are tailored to **specific modality combinations** (e.g., **text-image** or **audio-video**), rather than a **unified system** that can simultaneously analyze **text, images, videos, and audio**.
- Existing approaches mainly focus on **high benchmark accuracy**. **Real-world deployment**, **computational efficiency**, and **lightweight web-based implementation** have been given comparatively less attention.
- Many detection models do not **generalize** well **across datasets** and suffer performance drops when tested on **unseen deepfake generation methods**, **compressed multimedia**, or **content from social media** in the wild.
- Although **multimodal feature fusion** has boosted the detection performance, many existing methods rely on **complex architectures** that are computationally expensive, limiting their practical applications.
- Most existing systems are **black-box models**, that only generate **binary Real/Fake predictions** with poor **confidence estimation**, **interpretability**, or **explainable reasoning**.
- While most current research has focused on improving **individual detection models**, there are relatively few studies that have built **integrated end-to-end platforms** that incorporate **multimodal detection**, **authenticity reporting**, and **intuitive user interfaces**.
- Most commercial authenticity verification solutions are **proprietary**, **subscription-based**, or provide **limited transparency**, thus less accessible to researchers, students, and educational institutions.

Proposed Research Contribution

The proposed **CheckReality AI** system contributes to the following to address the identified research gaps:

- **Develops a** single web-based platform for simultaneous analysis **text, images, videos, and audio** in a unified **multimodal framework.**
- **Integrates modality-specific deep learning models** with an efficient **multimodal feature fusion mechanism** to produce a holistic authenticity assessment.

- **It does confidence-aware decision-making** and gives a score for overall authenticity as well as prediction confidence for each modality.
- **Generates explainable authenticity reports** to improve transparency and interpretability of AI predictions.
- **Develops a lightweight, scalable, and modular architecture** ready for practical deployment in educational, research, media, and cybersecurity applications.
- **Offers a flexible framework** that can be extended with future multimodal AI models, benchmark datasets, and emerging deepfake detection techniques.

Thus, the proposed research contributes toward developing a practical, scalable, explainable, and unified multimodal digital content authenticity verification system overcoming several limitations identified in recent literature, while supporting efficient deployment in real-world environments.

5. OBJECTIVES

The main objective of the proposed research is to develop a **unified, explainable, and multimodal Artificial Intelligence-based system** for the verification of the authenticity of digital content, by analyzing simultaneously **text, images, videos, and audio** within a single detection framework.

The specific objectives of the proposed system are:

- **Develop deep learning modules** for detecting **AI-generated text**, **deepfake images**, **manipulated videos**, and **synthetic speech** in a **modality-specific fashion**.
- **To design a cross-modal feature fusion framework** to exploit information from **text, image, video, and audio** modalities to generate a **unified and robust authenticity assessment.**
- **To implement an explainable and confidence-aware decision framework** to improve **reliability, transparency, and interpretability** of multimodal authenticity verification.
- **To evaluate and compare the proposed multimodal framework with existing unimodal and multimodal approaches, using standard benchmark datasets and performance indicators in terms of accuracy, robustness, computational efficiency, and practical applicability.**

6. METHODOLOGY / PLANNING OF WORK

The **CheckReality AI** system is a **modular multimodal Artificial Intelligence framework** to detect **AI-generated text**, **deepfake images**, **manipulated videos**, and **synthetic speech**. The methodology consists of the following steps.

Step 1: Dataset Collection

For each media modality, the public benchmark datasets will be collected.

Modality	Dataset
Text	Fake & Real News Dataset, AI vs Human Text Dataset
Image	140K Real & Fake Faces Dataset
Video	FaceForensics++ Dataset
Audio	ASVspoof 2019 Dataset

The datasets will be split into a **training**, a **validation**, and a **test** subset for model development and evaluation.

Step 2: Data Preprocessing

Modality-specific preprocessing techniques will be applied to improve data quality before model training.

- **Text:** Cleaning, tokenization, normalization, and transformer encoding.
- **Image:** Face detection, resizing, normalization, and augmentation.
- **Video:** Frame extraction, face detection, resizing, and temporal sequence generation.
- **Audio:** Noise reduction, normalization, Mel-spectrogram generation, and feature extraction.

Step 3: Feature Extraction and Modality-Specific Encoding

We will fine-tune pre-trained deep learning models to extract high-level semantic features from each modality.

Modality	Model
Text	RoBERTa
Image	EfficientNet
Video	EfficientNet + LSTM
Audio	CNN (Mel-Spectrogram)

The extracted feature embeddings will be fed to the multimodal fusion module.

Step 4: Cross-Modal Feature Fusion

The feature representations of each modality will be combined via a **cross-modal feature fusion framework**. The fusion module will integrate complementary information from **text, image, video and audio** to generate a unified multimodal representation for verifying authenticity.

Step 5: Attention-Based Decision and Explainability

An adaptive **attention-based decision module** will adaptively assign the importance of different modalities according to their relevance. The framework will also involve **confidence calibration** and **Explainable Artificial Intelligence (XAI)** to generate trustworthy predictions t with confidence scores and interpretable authenticity reports.

Step 6: Authenticity Prediction and Web Deployment

The integrated multimodal representation will produce the final **Real/Fake** prediction, an **overall authenticity score**, **modality-wise confidence**, and an **authenticity report**. We are going to build a **web application based on FastAPI.** Users will be able to upload text, images, videos, and audio for authenticity checking.

Step 7: Performance Evaluation

The proposed framework will be evaluated on standard benchmark datasets and compared with existing unimodal and multimodal approaches.

The evaluation will be conducted by:

- **Accuracy**
- **Precision**
- **Recall**
- **F1-score**
- **ROC-AUC**
- **Inference Time**
- **Computational Efficiency**
- **Cross-dataset Generalization**

The obtained results will be analyzed to assess the **accuracy**, **robustness**, **computational efficiency**, and **practical applicability** of the proposed multimodal framework.

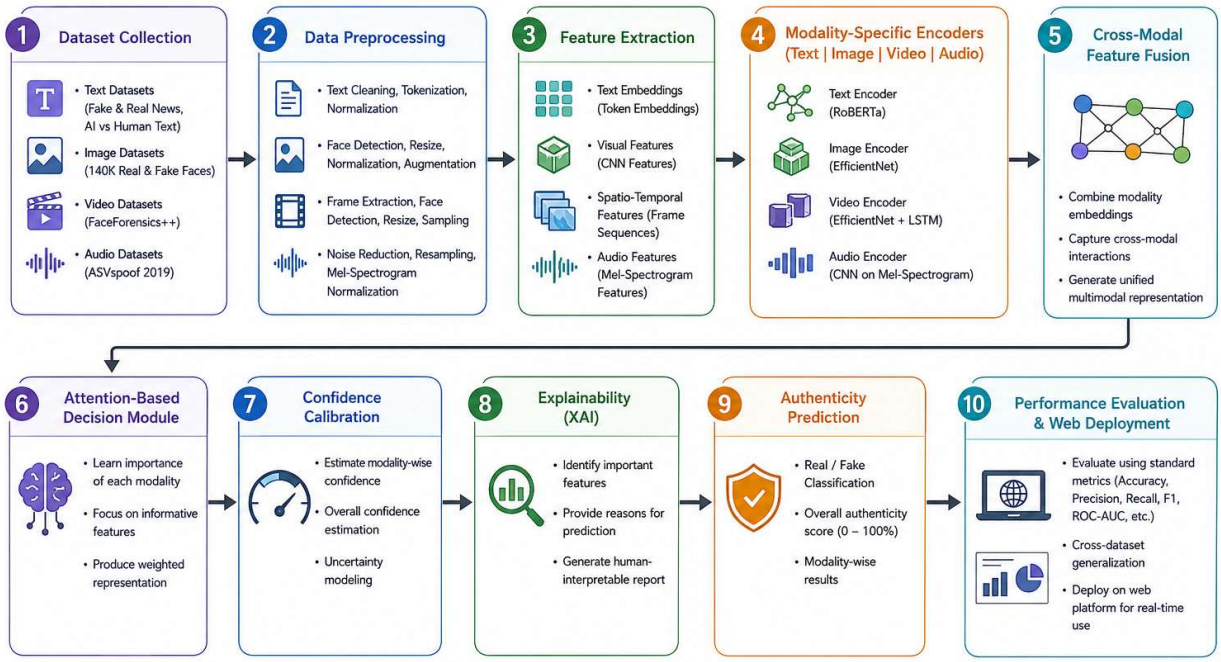


Figure 1: *Methodology Workflow of CheckReality AI: A Multimodal Deepfake and Misinformation Detection System for Digital Content Authenticity.*

7. FACILITIES REQUIRED FOR PROPOSED WORK

Software Requirements

Component	Technology / Tool
Programming Language	Python 3.11+
Backend Framework	FastAPI
Frontend	HTML5, CSS3, JavaScript, Bootstrap
Deep Learning Framework	PyTorch
Natural Language Processing	Hugging Face Transformers, NLTK
Computer Vision	OpenCV, EfficientNet
Video Processing	OpenCV, FFmpeg
Audio Processing	Librosa
Machine Learning Utilities	Scikit-learn, NumPy, Pandas
Database	MongoDB
Model Explainability (XAI)	SHAP / Grad-CAM *(or equivalent XAI library)*
Version Control	Git & GitHub
Development Environment	Visual Studio Code
Operating System	Windows 11 / Ubuntu Linux

Hardware Requirements

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i5 (11th Gen) / AMD Ryzen 5	Intel Core i7 / AMD Ryzen 7
RAM	16 GB	32 GB
Storage	512 GB SSD	1 TB NVMe SSD
GPU	NVIDIA GTX 1650 / RTX 2050 (4 GB VRAM)	NVIDIA RTX 3060 / RTX 4060 or higher (8 GB+ VRAM)
Operating System	Windows 11 / Ubuntu Linux	Windows 11 / Ubuntu Linux

Internet & Cloud Requirements

- **Stable broadband Internet connection** for downloading pretrained models and benchmark datasets.
- Use **GitHub** for version control and collaboration.
- **Google Colab / Kaggle Notebook (optional)** for model training when local GPU resources are limited.

Benchmark Datasets

Dataset	Media	Approx. Size	Classes	Why Used
Fake & Real News (Kaggle)	Text	~44k articles	Fake / Real	Fake news detection
AI vs Human Text	Text	~487k+ samples	AI / Human	AI-generated text detection
FaceForensics++	Video	~1,000 original videos + manipulated versions	Real / Fake	Deepfake video detection
140k Real & Fake Faces	Image	~140,000 images	Real / Fake	Face forgery detection
ASVspoof 2019	Audio	~250k+ utterances across train/dev/eval	Bonafide / Spoof	Synthetic/cloned voice detection

Pretrained models

Task	Pretrained Model	Fine-tune On
Text	**RoBERTa-base**	Fake News + AI vs Human Text
Image	**EfficientNet-B0/B3**	140k Real/Fake Faces
Video	**EfficientNet + LSTM**	FaceForensics++
Audio	**CNN** (trained on Mel Spectrograms)	ASVspoof 2019

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Dataset References:

Text

Kaggle Fake & Real News Dataset

AI vs Human Text Dataset

Video

FaceForensics++ Dataset

Image

140k Real and Fake Faces Dataset

Audio

ASVspoof 2019 Official Dataset

ASVspoof 2019 Kaggle Dataset